

SENSOR BASED SIMULATION USING TINKERCAD / PROTEUS

Automatic LED ON/OFF Using LDR Sensor

Aim

To simulate an IoT-based automatic lighting system using an LDR sensor and Arduino in Tinkercad/Proteus where the LED turns ON in darkness and OFF in light.

Objective

- To understand sensor-based automation.
- To learn Arduino interfacing with sensors.
- To simulate IoT systems using Tinkercad/Proteus.
- To control an LED automatically using light intensity.

Components Required

S.No	Component	Quantity
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1	Arduino Uno	1
2	Breadboard	1
3	LDR Sensor	1
4	LED	1
5	220 Ω Resistor	1
6	10k Ω Resistor	1
7	Jumper Wires	Few

Software Required

- Tinkercad
- Proteus Design Suite
- Arduino IDE

Theory

An LDR (Light Dependent Resistor) is a sensor whose resistance changes according to the intensity of light.

- In darkness → resistance increases.
- In bright light → resistance decreases.

The Arduino reads the analog value from the LDR sensor and controls the LED accordingly.

If the environment is dark:

- LED turns ON.

If the environment is bright:

- LED turns OFF.

This concept is widely used in:

- Automatic street lights
- Smart homes
- Energy-saving systems
- Security lighting systems

Circuit Connections

LED Connection

LED Pin	Arduino Connection
Positive (+)	Digital Pin 13
Negative (-)	GND through 220Ω resistor

LDR Connection

LDR Connection	Arduino Connection
One terminal	5V
Other terminal	Analog Pin A0
10kΩ resistor	Connected between A0 and GND

Circuit Diagram Description

1. Connect the LED positive terminal to pin 13.
2. Connect LED negative terminal to GND using a 220 Ω resistor.
3. Connect one side of LDR to 5V.
4. Connect the second side of LDR to A0.
5. Connect a 10k Ω resistor between A0 and GND.
6. Upload the Arduino program.
7. Start the simulation.

Arduino Program

```
int ldr = A0;
```

```
int led = 13;
```

```
int value = 0;
```

```
void setup()
```

```
{
```

```
pinMode(led, OUTPUT);
```

```
Serial.begin(9600);
```

```
}
```

```
void loop()
```

```
{
```

```
value = analogRead(ldr);
```

```
Serial.println(value);
```

```
if(value < 500)
```

```
{
```

```
digitalWrite(led, HIGH);
```

```
}
```

```

else
{
digitalWrite(led, LOW);
}

delay(100);
}

```

Code Explanation

Variable Declaration

```
int ldr = A0;
```

- Connects the LDR sensor to Analog Pin A0.

```
int led = 13;
```

- Connects the LED to Digital Pin 13.

```
int value = 0;
```

- Stores the sensor reading value.

Setup Function

```
void setup()
```

- Runs only once when Arduino starts.

```
pinMode(led, OUTPUT);
```

- Sets LED pin as output.

```
Serial.begin(9600);
```

- Starts serial communication.

Loop Function

```
value = analogRead(ldr);
```

- Reads the analog value from the LDR.

```
if(value < 500)
```

- Checks whether the light intensity is low.

`digitalWrite(led, HIGH);`

- Turns ON the LED.

`digitalWrite(led, LOW);`

- Turns OFF the LED.

`delay(100);`

- Waits for 100 milliseconds.

Simulation Steps in Tinkercad

1. Open Tinkercad.
2. Click on “Create New Circuit”.
3. Add Arduino Uno.
4. Add LED, LDR, and resistors.
5. Make the circuit connections.
6. Copy and paste the Arduino code.
7. Click “Start Simulation”.
8. Adjust the LDR light intensity.
9. Observe LED turning ON/OFF automatically.

Simulation Steps in Proteus

1. Open Proteus Design Suite.
2. Create a new project.
3. Add Arduino Uno.
4. Add LED and LDR components.
5. Connect all components.
6. Compile Arduino code using Arduino IDE.
7. Generate HEX file.
8. Upload HEX file to Arduino in Proteus.
9. Run the simulation.
10. Observe the LED behaviour.

Expected Output

SENSOR BASED IoT SIMULATION LED ON/OFF USING LDR SENSOR

1. CIRCUIT DIAGRAM (Tinkercad)

LED → Pin 13
LDR → 5V and A0
10kΩ Resistor → A0 and GND
220Ω Resistor → Pin 13 and GND

2. TINKERCAD SIMULATION SCREENSHOTS

In Light (LED OFF)

LDR Value: 798

In Darkness (LED ON)

LDR Value: 134

3. PROTEUS SIMULATION SCREENSHOTS

In Light (LED OFF)

In Darkness (LED ON)

4. ARDUINO CODE

```
int ldr = A0;
int led = 13;
int value = 0;

void setup()
{
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  value = analogRead(ldr);
  Serial.println(value);
  if(value < 500)
  {
    digitalWrite(led, HIGH); // LED ON in darkness
  }
  else
  {
    digitalWrite(led, LOW); // LED OFF in light
  }
  delay(100);
}
```

5. CODE EXPLANATION

- ldr = A0 → LDR is connected to analog pin A0.
- led = 13 → LED is connected to digital pin 13.
- analogRead(ldr) reads the light intensity.
- In darkness, LDR resistance increases → value decreases.
- If value < 500 → it is dark → LED turns ON.
- If value ≥ 500 → it is bright → LED turns OFF.
- Serial.println(value) shows the LDR value in Serial Monitor.

6. COMPONENTS REQUIRED

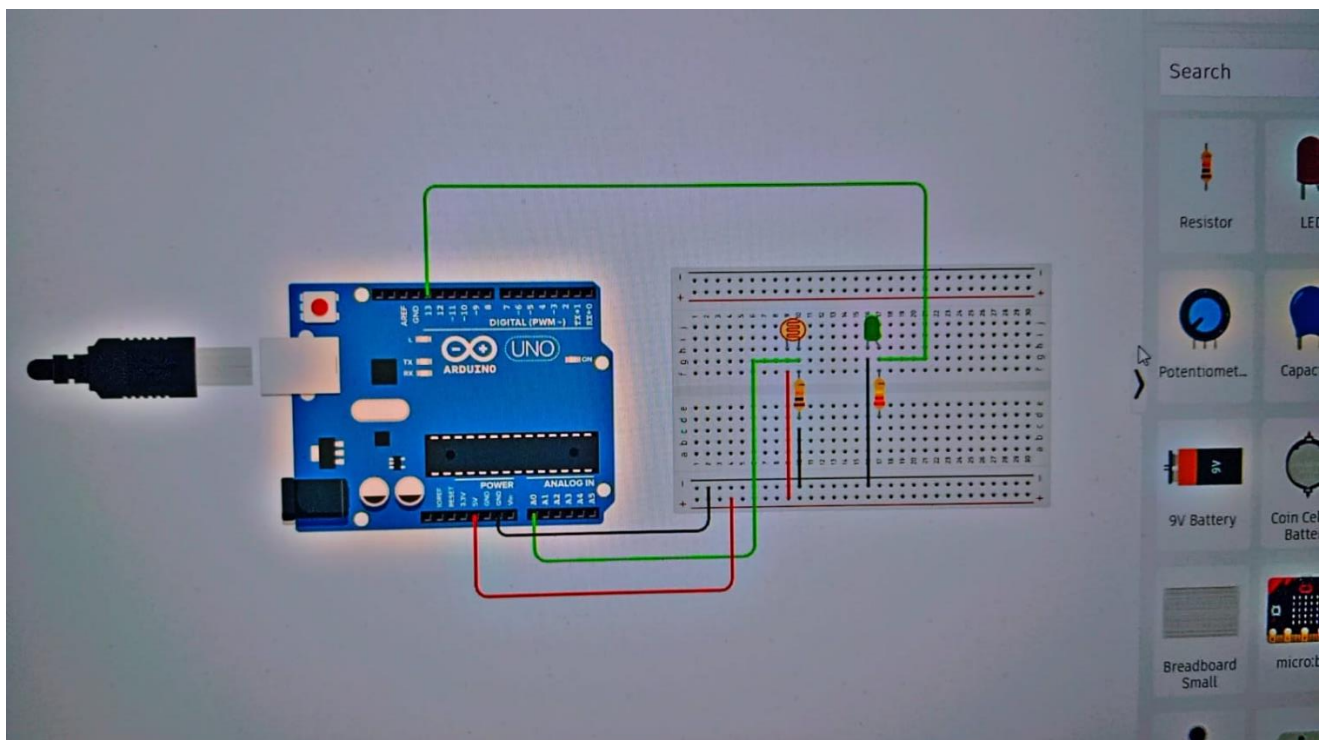
- Arduino Uno
- LDR Sensor
- LED
- 220Ω Resistor
- 10kΩ Resistor
- Breadboard and Jumper wires

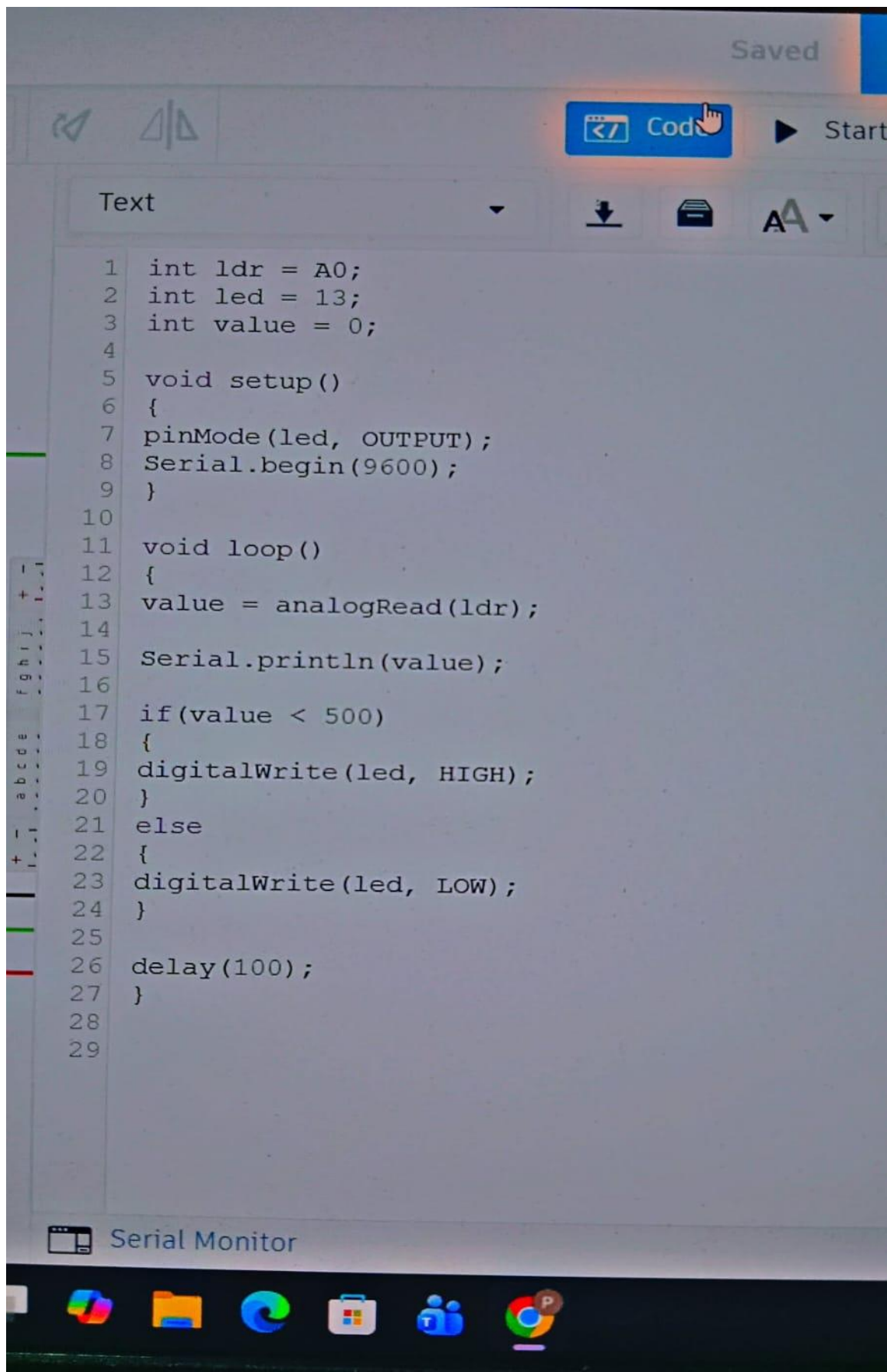
7. SERIAL MONITOR OUTPUT (Tinkercad/Arduino IDE)

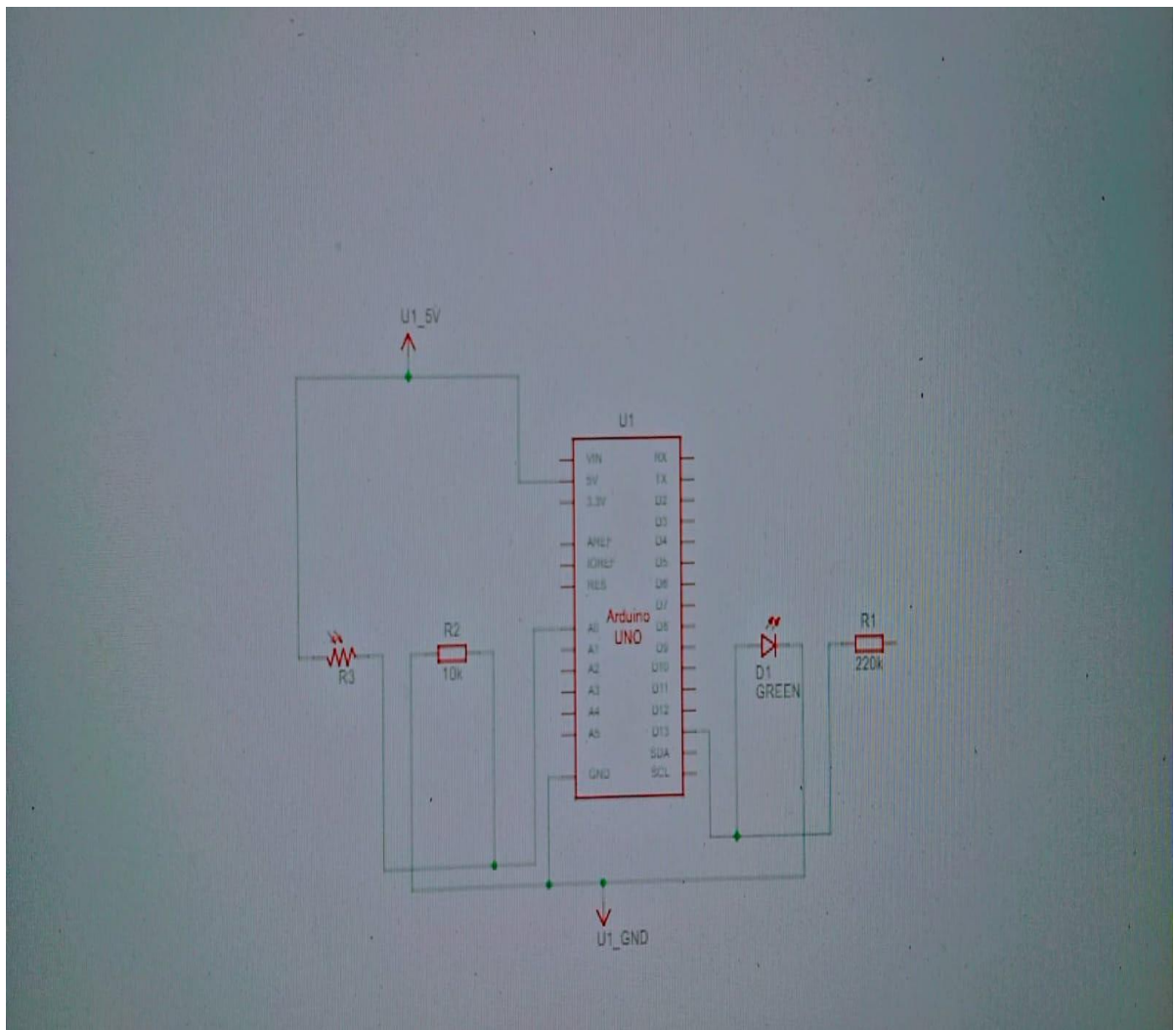
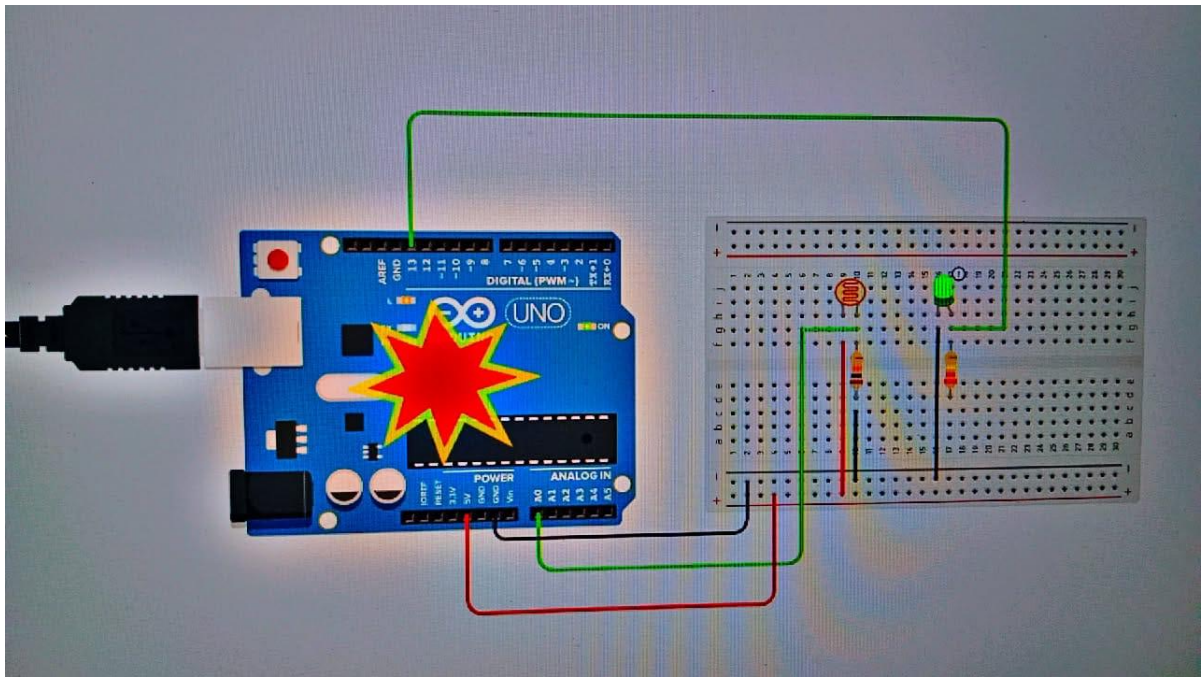
In Light

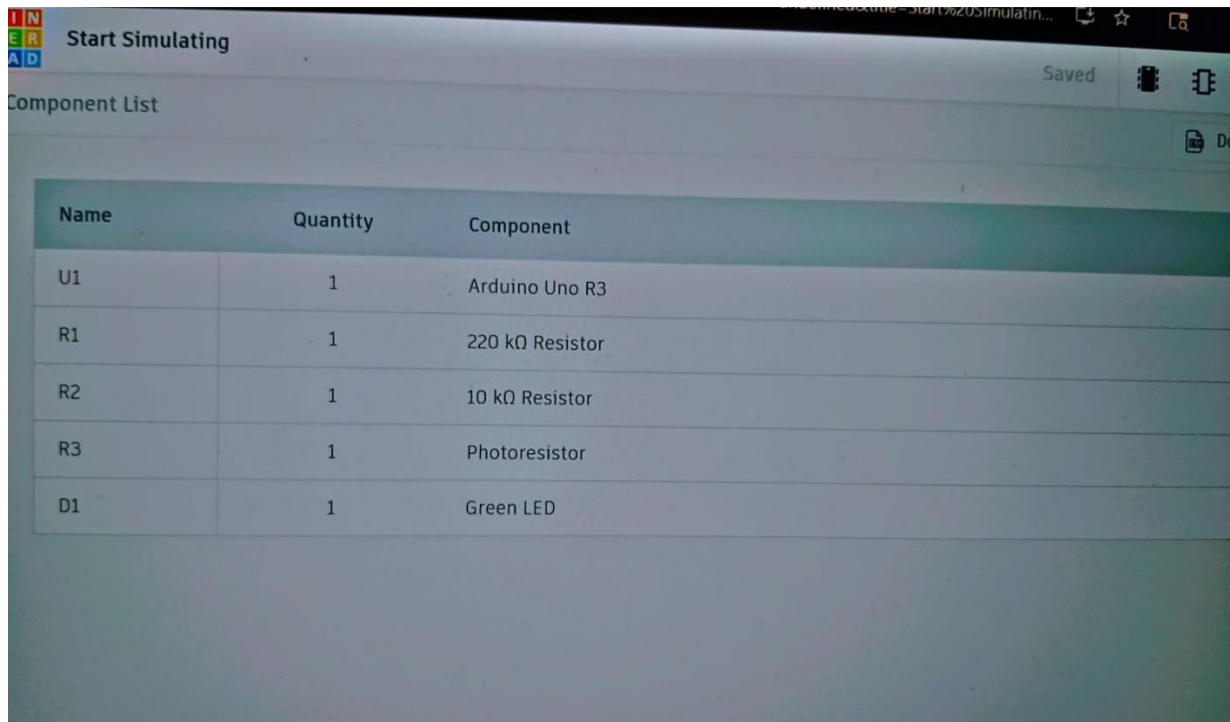
In Darkness

WORKING: The LDR senses the light intensity. In darkness, Arduino turns ON the LED. In light, Arduino turns OFF the LED.









Condition	LED Status
Dark Environment	ON
Bright Environment	OFF

Applications

- Automatic street lighting
- Smart home systems
- Industrial automation
- Security systems
- Energy conservation systems

Advantages

- Low power consumption
- Automatic operation
- Simple design

- Cost-effective
- Easy implementation

Conclusion

The sensor-based IoT simulation using Arduino and LDR was successfully implemented in Tinkercad/Proteus. The LED automatically turns ON in darkness and OFF in bright light based on the sensor input. This project demonstrates the working principle of automated lighting systems used in real-life IoT applications.